SEMESTER MAJOR ASSIGNMENT

Preprocessing Order, Regularization, & Imbalance Handling in EEG Seizure Prediction

Investigating Generalization Performance in Machine Learning

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BSCS-B-6th | Semester VI | Machine Learning
Institute of Management Sciences (IMSciences), Peshawar | 2024

DATASET COLLECTION & JUSTIFICATION

UCI Bonn

Controlled EEG segments. Ideal for testing baseline filters and stationary signal behavior.

Imbalance: 1:4

CHB-MIT

Continuous clinical Scalp EEG. High complexity and multi-channel dynamics.

Imbalance: 1:50 (Severe)

Siena Dataset

Adult Scalp EEG with significant noise and artifacts. Tests model robustness.

Imbalance: 1:5.7

Justification: Mimics real-world rarity of ictal events while providing thousands of hours of clinical recording.

PREPROCESSING ARCHITECTURAL FLOWS

Pipeline A: Signal-Centric

Standardization → Noise Removal → Univariate Feature Selection (SelectKBest)

Pipeline B: Feature-Centric

Feature Extraction → Robust Scaling → PCA (95% Variance)

Research Insight (Pipeline C): The Hybrid Model

Applying **PCA before Feature Selection** captures spatial cross-channel correlations that discrete selection often discards.

BASELINE MODEL: LOGISTIC REGRESSION

Used as the core estimator due to its interpretability and computational efficiency.

$$P(y=1 \mid x) = \frac{1}{1+e^{-(\beta_0+\beta^T x)}}$$

- **Accuracy:** General correctness of model.
- **F1-Score:** Balance of Precision and Recall.
- **PR-AUC:** Crucial for severe imbalance handling.

UNDERFITTING VS. OVERFITTING ANALYSIS

Scenario 1: Underfitting

Cause: Extreme L2 Regularization ($\lambda > 1000$) + Minimal Features.

Effect: High error on both training and validation sets. Decision boundary is too simple for complex EEG.

Scenario 2: Overfitting

Cause: No Regularization + High-dimensional Raw Data.

Effect: Near-zero training error but massive validation loss. Model captures signal noise.

Validation curves are essential to identify the "Sweet Spot" for maximum generalization.

REGULARIZATION STUDY & SPARSITY

L1 (Lasso)

Induces Sparsity. Automatically selects focal channels by zeroing out non-essential electrode weights.

$$J(W, b) = \frac{1}{m} \sum L + \frac{\lambda}{m} \| W \|_1$$

L2 (Ridge)

Provides Stability. Shrinks weights uniformly to handle highly correlated electrode data effectively.

$$J(W, b) = \frac{1}{m} \sum L + \frac{\lambda}{2m} \| W \|_2^2$$

HANDLING CLASS IMBALANCE

SMOTE / Oversampling:
Creates synthetic seizure samples in feature space.
Risk: Can introduce artifact-driven noise.

Class Weighting:
Heavily penalizes minority class misclassifications.
Benefit: Maintains original signal integrity.

PR-AUC Focus:
Evaluates Precision vs. Recall tradeoff without being fooled by high majority-class accuracy.

COMPARATIVE ANALYSIS RESULTS

Investigation Area	Research Findings	Primary Benefit
Preprocessing Order	Noise removal MUST precede scaling	Prevents artifact-driven variance inflation
Top Generalizer	L2 Regularization (Ridge)	Stable weight distribution across patients
Elastic Net Usage	Best for Feature-Extracted Pipelines	Addresses high feature collinearity
Interaction Strategy	Class Weighting + L2 Regularization	Most stable boundaries for clinical data

CONCLUSION & RECOMMENDATIONS

- **Hybrid Priority:** Focus on pipelines that account for spatial covariance (PCA + SelectKBest).
- **Diagnostic Metrics:** Always prioritize PR-AUC over Accuracy for medical tasks.
- **Signal Stability:** Class Weighting is statistically safer than SMOTE for sensitive biomedical signals.
- **Global Generalization:** L2 Regularization is essential for cross-dataset performance stability.

Questions?

Thank you for your attention.

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IMAGE SOURCES

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